Appendix – Relative Value with the Gauss+ model for the yield curve

This appendix explicitly derives the model solution sticking to the notation in Tuckman and Serrat (2022). All expectation operators and brownian motions are defined under the $\mathbb{Q}$ measure unless otherwise specified.

Solving the model

We start from the three equations, where $\alpha_r, \alpha_m, \alpha_l > 0$:

$$dr_t = -\alpha_r(r_t - m_t)dt$$

$$dm_t = -\alpha_m(m_t - l_t)dt + \sigma_m\left(\rho dW_t^1 + \sqrt{1-\rho^2}dW_t^2\right)$$

$$dl_t = -\alpha_l(l_t - \mu)dt + \sigma_l dW_t^1$$

Where the brownian motions are specified under the risk-neutral measure. Expanding the drift terms:

$$dr_t = (-\alpha_r r_t + \alpha_r m_t)dt$$

$$dm_t = (-\alpha_m m_t + \alpha_m l_t)dt + \sigma_m\left(\rho dW_t^1 + \sqrt{1-\rho^2}dW_t^2\right)$$

$$dl_t = (-\alpha_l l_t + \alpha_l \mu)dt + \sigma_l dW_t^1$$

Now let $x_t = (r_t, m_t, l_t)$ and note how (signs chosen to follow the book):

$$\underbrace{\begin{pmatrix} dr_t \\ dm_t \\ dl_t \end{pmatrix}}_{dx_t} = \underbrace{\begin{bmatrix} -\alpha_r & +\alpha_r & 0 \\ 0 & -\alpha_m & +\alpha_m \\ 0 & 0 & -\alpha_l \end{bmatrix}}_{R(\alpha)} \underbrace{\begin{pmatrix} r_t \\ m_t \\ l_t \end{pmatrix}}_{x_t} dt + \cdots$$

Where $\alpha = (\alpha_r, \alpha_m, \alpha_l)$. If we further define $\bar{\mu} = (\mu, \mu, \mu)$:

$$dx_t = R(\alpha)x_t dt + \underbrace{\begin{pmatrix} 0 \\ 0 \\ +\alpha_l\mu \end{pmatrix}}_{-R(\alpha)\bar{\mu}} dt + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ \rho\sigma_m & \sqrt{1-\rho^2}\sigma_m & 0 \\ \sigma_l & 0 & 0 \end{bmatrix}}_{\Omega(\sigma)} \underbrace{\begin{pmatrix} 0 \\ dW_t^1 \\ dW_t^2 \end{pmatrix}}_{dW_t}$$

Where $\sigma = (\sigma_m, \sigma_l, \rho)$. Which brings us to a clean matrix formulation:

$$dx_t = R(\alpha)[x_t - \bar{\mu}]dt + \Omega(\sigma)dW_t$$

Now the problem is that $R(\alpha)$ is not diagonal, which means the model is still in cascade form. We want to transform it so as to derive a form with three factors that move independently of each other. Assume we want new factors X_t that are related to x_t in this way, for some A:

$$x_t - \bar{\mu} = AX_t$$

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If this is true, then also:

$$dx_t = AdX_t$$

And substituting into the cascade form equation:

$$AdX_t = R(\alpha)AX_t dt + \Omega(\sigma)dW_t$$

$$dX_t = [A^{-1}R(\alpha)A]X_t dt + A^{-1}\Omega(\sigma)dW_t$$

A convenient choice for A would be that of the eigenvectors of $R(\alpha)$. This makes the matrix multiplying X_t diagonal. Recall that the eigenvalues of $R(\alpha)$ are the elements on its diagonal. Hence $A = A(\alpha)$ satisfies:

$$R(\alpha)A(\alpha) = A(\alpha)\begin{bmatrix} -\alpha_r & 0 & 0 \\ 0 & -\alpha_m & 0 \\ 0 & 0 & -\alpha_l \end{bmatrix}$$

Which leads us to the reduced form:

$$dX_t = -\mathrm{diag}(\alpha)X_t dt + A^{-1}(\alpha)\Omega(\sigma)dW_t$$

Rewriting the link between x_t and X_t gives:

$$x_t = A(\alpha)X_t + \bar{\mu}$$

It can be shown that $A(\alpha)$ takes the form:

$$A(\alpha) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & \dfrac{\alpha_r - \alpha_m}{\alpha_r} & \dfrac{\alpha_r - \alpha_l}{\alpha_r} \\ 0 & 0 & \dfrac{(\alpha_r - \alpha_l)(\alpha_m - \alpha_l)}{\alpha_r \alpha_m} \end{bmatrix}$$

Reading this statement only in the first row:

$$r_t = 1'X_t + \mu$$

By no-arbitrage, the price of a zero-coupon bond with maturity τ at time t is given by:

$$P_t(\tau) = E_t^Q\left[e^{-\int_t^{t+\tau} r_s ds}\right]$$

Which, incorporating the relationship with X_t and μ, becomes:

$$P_t(\tau) = E_t\left[\exp\left\{-\int_t^{t+\tau}(1'X_s + \mu)\,ds\right\}\right]$$

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Where the expectation is taken with respect to the risk-neutral measure. We need to figure out a way to rewrite this in terms of the parameters. The reasoning is: if we can prove that X_t is gaussian, then also r_s is, and the random variable inside the expectation is lognormally distributed. If we know the mean and variance characterizing the distribution of X_t then we can solve for the expectation of $e^{-\int_t^{t+\tau} r_s ds}$.

First of all, we solve the SDE governing X_t, so we can show that it's normally distributed and compute its mean and variance.

$$dX_t = -\mathrm{diag}(\alpha)X_t dt + A^{-1}(\alpha)\Omega(\sigma)dW_t$$

With the integrating factor $e^{\mathrm{diag}(\alpha)t} = \mathrm{diag}(e^{\alpha_r t}, e^{\alpha_m t}, e^{\alpha_l t})$:

$$e^{\mathrm{diag}(\alpha)t} \cdot dX_t + e^{\mathrm{diag}(\alpha)t}\mathrm{diag}(\alpha)X_t dt = e^{\mathrm{diag}(\alpha)t} \cdot A^{-1}(\alpha)\Omega(\sigma)dW_t$$

Notice how the left hand side is the differential of $e^{\mathrm{diag}(\alpha)t}X_t$:

$$d\left(e^{\mathrm{diag}(\alpha)t}X_t\right) = e^{\mathrm{diag}(\alpha)t} \cdot dX_t + d\left(e^{\mathrm{diag}(\alpha)t}\right)X_t dt$$

$$d\left(e^{\mathrm{diag}(\alpha)t}X_t\right) = e^{\mathrm{diag}(\alpha)t} \cdot dX_t + e^{\mathrm{diag}(\alpha)t}\mathrm{diag}(\alpha)X_t dt$$

So we can rewrite the expression as:

$$d(e^{\mathrm{diag}(\alpha)t} \cdot X_t) = e^{\mathrm{diag}(\alpha)t} \cdot A^{-1}(\alpha)\Omega(\sigma)dW_t$$

Integrating between t and $t+\tau$ on both sides we get:

$$e^{\mathrm{diag}(\alpha)(t+\tau)}X_{t+\tau} - e^{\mathrm{diag}(\alpha)t}X_t = \int_t^{t+\tau} e^{\mathrm{diag}(\alpha)s} \cdot A^{-1}(\alpha)\Omega(\sigma)dW_s$$

And solving for $X_{t+\tau}$ we get:

$$X_{t+\tau} = e^{-\mathrm{diag}(\alpha)\tau}X_t + \int_t^{t+\tau} e^{-\mathrm{diag}(\alpha)(t+\tau-s)} \cdot A^{-1}(\alpha)\Omega(\sigma)dW_s$$

If we have a starting value X_t that is a constant, then all the randomness in $X_{t+\tau}$ comes from the stochastic integral. Since the stochastic integral is a linear function of the brownian increment vectors dW_s then the integral follows a normal distribution. And thus also $X_{t+\tau}$ follows a normal. Since the stochastic integral has zero mean, we easily compute the expectation of $X_{t+\tau}$:

$$E_t[X_{t+\tau}] = e^{-\mathrm{diag}(\alpha)\tau}X_t$$

By ito isometry $V\left(\int H_s dW_s\right) = \int H_s H_s' ds$, we can compute its variance:

$$V_t(X_{t+\tau}) = V_t\left(\int_t^{t+\tau} e^{-\mathrm{diag}(\alpha)(t+\tau-s)} \cdot A^{-1}(\alpha)\Omega(\sigma)dW_s\right)$$

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$$V_t(X_{t+\tau}) = \int_t^{t+\tau} \left[e^{-\mathrm{diag}(\alpha)(t+\tau-s)} \cdot A^{-1}(\alpha)\Omega(\sigma)\right]\left[e^{-\mathrm{diag}(\alpha)(t+\tau-s)} \cdot A^{-1}(\alpha)\Omega(\sigma)\right]' ds$$

$$V_t(X_{t+\tau}) = \int_t^{t+\tau} e^{-\mathrm{diag}(\alpha)(t+\tau-s)}[A^{-1}(\alpha)\Omega(\sigma)][\Omega(\sigma)'A^{-1}(\alpha)']e^{-\mathrm{diag}(\alpha)(t+\tau-s)} ds$$

Taking out terms that do not depend on the variable with respect to which we're integrating:

$$V_t(X_{t+\tau}) = [A^{-1}(\alpha)\Omega(\sigma)][\Omega(\sigma)'A^{-1}(\alpha)'] \int_t^{t+\tau} e^{-\mathrm{diag}(\alpha)(t+\tau-s)} e^{-\mathrm{diag}(\alpha)(t+\tau-s)} ds$$

Now, the exponential matrices are diagonal, so also $V_t(X_{t+\tau})$ will be. Hence entrywise this is:

$$[V_t(X_{t+\tau})]_{ij} = \{[A^{-1}(\alpha)\Omega(\sigma)][\Omega(\sigma)'A^{-1}(\alpha)']\}_{ij} \int_t^{t+\tau} e^{-(\alpha_i+\alpha_j)(t+\tau-s)} ds$$

The integral can be solved easily with a change of variable $u = t + \tau - s$

$$\int_t^{t+\tau} e^{-(\alpha_i+\alpha_j)(t+\tau-s)} ds = \int_0^{\tau} e^{-(\alpha_i+\alpha_j)u}\, du = \left[\frac{e^{-(\alpha_i+\alpha_j)u}}{-(\alpha_i+\alpha_j)}\right]_0^{\tau} = \frac{e^{-(\alpha_i+\alpha_j)\tau}}{-(\alpha_i+\alpha_j)} + \frac{1}{-(\alpha_i+\alpha_j)} = \frac{1-e^{-(\alpha_i+\alpha_j)\tau}}{\alpha_i+\alpha_j)}$$

Which leads to the formulation:

$$[V_t(X_{t+\tau})]_{ij} = \{[A^{-1}(\alpha)\Omega(\sigma)][\Omega(\sigma)'A^{-1}(\alpha)']\}_{ij} \cdot \frac{1-e^{-(\alpha_i+\alpha_j)\tau}}{\alpha_i+\alpha_j)}$$

Ok. So let's go back to the formulation of $P_t(\tau)$ in terms of X_t. We simplify it a bit:

$$P_t(\tau) = E_t\left[\exp\left\{\left(-\int_t^{t+\tau} 1'X_s\, ds\right) + \mu\tau\right\}\right]$$

The integral $\int_t^{t+\tau} 1'X_s\, ds$ is a linear function of X_s, which as we said follows a normal conditional to X_t for all $s > t$. Hence the integral follows a univariate normal, with a certain mean m and variance v^2. If we can compute m and v, we then know that:

$$P_t(\tau) = E_t\left[\exp\left\{\left(-\int_t^{t+\tau} 1'X_s\, ds\right)\right\}\right] e^{-\mu\tau} = e^{-\mu\tau - m + \frac{1}{2}v^2}$$

Ok. Now let's compute m and v^2. We start with m:

$$m = E_t\left[\left(\int_t^{t+\tau} 1'X_s\, ds\right)\right] = \int_t^{t+\tau} 1'E_t[X_s]\, ds$$

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Since $|1'X_s|$ has finite expectation, we can swap integration and summation by Fubini's theorem Then by substituting in the formula we derived for $E_t[X_s]$ and with a change of variable $u = s - t$ we get:

$$m = \int_t^{t+\tau} 1' e^{-\mathrm{diag}(\alpha)(s-t)} X_t \, ds = X_t \int_0^{\tau} 1' e^{-\mathrm{diag}(\alpha)u} \, du$$

$$m = \sum_{i=1}^{3} \left(X_t^i \int_0^t e^{-\alpha_i u} du \right) = \sum_{i=1}^{3} \left(X_t^i \frac{1 - e^{-\alpha_i \tau}}{\alpha_i} \right)$$

Now define a vector $\tilde{B}(\tau, \alpha)$ of the form:

$$\tilde{B}(\tau, \alpha) := \begin{pmatrix} \frac{1 - e^{-\alpha_r \tau}}{\alpha_r} & \frac{1 - e^{-\alpha_m \tau}}{\alpha_m} & \frac{1 - e^{-\alpha_l \tau}}{\alpha_l} \end{pmatrix}'$$

Then m can be conveniently rewritten as:

$$m(\tau, \alpha) = \tilde{B}(\tau, \alpha) X_t$$

Now, to compute the variance of the integral, let's rewrite it with the solution of the SDE for X_t:

$$1' X_{t+\tau} = 1' \left(e^{-\mathrm{diag}(\alpha)\tau} X_t + \int_t^{t+\tau} e^{-\mathrm{diag}(\alpha)(t+\tau-s)} \cdot A^{-1}(\alpha) \Omega(\sigma) dW_s \right)$$

$$1' X_{t+\tau} = 1' e^{-\mathrm{diag}(\alpha)\tau} X_t + 1' \int_t^{t+\tau} e^{-\mathrm{diag}(\alpha)(t+\tau-s)} \cdot A^{-1}(\alpha) \Omega(\sigma) dW_s$$

The variance of the integral of $1'X_s$ only comes from the integral of the stochastic integral:

$$v^2 = V_t \left(\int_t^{t+\tau} 1' X_s \, ds \right) = V_t \left(\int_t^{t+\tau} \left(\int_t^{s} 1' e^{-\mathrm{diag}(\alpha)(s-u)} \cdot A^{-1}(\alpha) \Omega(\sigma) dW_u \right) ds \right)$$

Since the term $H(s, u) = 1' e^{-\mathrm{diag}(\alpha)(s-u)} \cdot A^{-1}(\alpha) \Omega(\sigma)$ is square integrable, by Fubini we can swap the orders of integration:

$$v^2 = V_t \left(\int_t^{t+\tau} \left(\int_u^{t+\tau} 1' e^{-\mathrm{diag}(\alpha)(s-u)} ds \right) \cdot A^{-1}(\alpha) \Omega(\sigma) dW_u \right)$$

Setting $z = s - u$ in the inner integral we get

$$\int_u^{t+\tau} 1' e^{-\mathrm{diag}(\alpha)(s-u)} ds = \int_0^{t+\tau-u} 1' e^{-\mathrm{diag}(\alpha)z} dz = \begin{pmatrix} \frac{1 - e^{\alpha_r (t+\tau-u)}}{\alpha_r} & \frac{1 - e^{\alpha_m (t+\tau-u)}}{\alpha_m} & \frac{1 - e^{\alpha_l (t+\tau-u)}}{\alpha_l} \end{pmatrix}' :$$
$$= g(t + \tau - u)$$

Plugging this new term into the variance of the other integral we get:

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$$v^2 = V_t\left(\int_t^{t+\tau} g(t+\tau-u)'A^{-1}(\alpha)\Omega(\sigma)\,dW_u\right)$$

We use Ito isometry and we obtain:

$$v^2 = \int_t^{t+\tau} [g(t+\tau-u)'A^{-1}(\alpha)\Omega(\sigma)][g(t+\tau-u)'A^{-1}(\alpha)\Omega(\sigma)]'\,du$$

Expanding componentwise the quadratic form inside the integral:

$$v^2 = \sum_{i,j}[A^{-1}(\alpha)\Omega(\sigma)\Omega(\sigma)'A^{-1}(\alpha)']_{ij}\int_t^{t+\tau} g(t+\tau-u)_i' g(t+\tau-u)_j\,du$$

The TS appendix defines $\sigma_{ij} = [A^{-1}(\alpha)\Omega(\sigma)\Omega(\sigma)'A^{-1}(\alpha)']_{ij}$ for ease of notation.

$$v^2 = \sum_{i,j}\left\{\sigma_{ij}\int_t^{t+\tau}\left[\frac{1-e^{\alpha_i(t+\tau-u)}}{\alpha_i}\right]\left[\frac{1-e^{\alpha_j(t+\tau-u)}}{\alpha_j}\right]du\right\}$$

By the change of variable $h = t+\tau-u$, and expanding the product of the two exponentials:

$$v^2 = \sum_{i,j}\left\{\frac{\sigma_{ij}}{\alpha_i\alpha_j}\int_0^{\tau}(1-e^{\alpha_i h})(1-e^{\alpha_j h})\,dh\right\}$$

$$v^2 = \sum_{i,j}\left\{\frac{\sigma_{ij}}{\alpha_i\alpha_j}\int_0^{\tau}1-e^{-\alpha_i h}-e^{-\alpha_j h}+e^{-(\alpha_i+\alpha_j)h}\,dh\right\}$$

$$v^2 = \sum_{i,j}\left\{\frac{\sigma_{ij}}{\alpha_i\alpha_j}\left[\tau-\frac{1-e^{-\alpha_i\tau}}{\alpha_i}-\frac{1-e^{-\alpha_j\tau}}{\alpha_j}+\frac{1-e^{-(\alpha_i+\alpha_j)\tau}}{\alpha_i+\alpha_j}\right]\right\}$$

Ok! So now we can plug m and v into the expression for the price of a ZCB. More cleanly, let's define the price of a ZCB in terms of its ZCB yield:

$$P_t(\tau) = \exp(-y_t(\tau)\tau)$$

$$y_t(\tau) = -\frac{1}{\tau}\log P_t(\tau)$$

Recall that $P_t(\tau) = e^{-\mu\tau-m+\frac{1}{2}v^2}$ so then:

$$y_t(\tau) = -\frac{1}{\tau}\left\{-\mu\tau-m+\frac{1}{2}v^2\right\}$$

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$$y_t(\tau) = \mu + \frac{m}{\tau} - \frac{1}{2\tau}v^2$$

Good. Now, in order to stick to the book's notation, define a new vector:

$$B(\tau,\alpha) = \frac{1}{\tau}\tilde{B}(\tau,\alpha) = \begin{pmatrix} \frac{1-e^{-\alpha_r\tau}}{\alpha_r\tau} & \frac{1-e^{-\alpha_m\tau}}{\alpha_m\tau} & \frac{1-e^{-\alpha_l\tau}}{\alpha_l\tau} \end{pmatrix}'$$

And also a new quantity $C(\tau,\alpha,\sigma)$ defined as:

$$\begin{aligned} C(\tau,\alpha,\sigma) &= \sum_{i=1}^{3}\sum_{j=1}^{3}\frac{\sigma_{ij}}{2\alpha_i\alpha_j}\left[1 - \frac{1-e^{-\alpha_i\tau}}{\alpha_i\tau} - \frac{1-e^{-\alpha_j\tau}}{\alpha_j\tau} + \frac{1-e^{-(\alpha_i+\alpha_j)\tau}}{(\alpha_i+\alpha_j)\tau}\right] \\ &= \sum_{i=1}^{3}\sum_{j=1}^{3}\frac{\sigma_{ij}}{2\alpha_i\alpha_j}\left[1 - B(\tau,\alpha)_i - B(\tau,\alpha)_j + \frac{1-e^{-(\alpha_i+\alpha_j)\tau}}{(\alpha_i+\alpha_j)\tau}\right] = \frac{1}{2\tau}v^2 \end{aligned}$$

Since $m = \tilde{B}(t,\alpha)X_t$ we finally get to the formula from the book:

$$y_t(\tau) = \mu - C(\tau,\alpha,\sigma) + B(\tau,\alpha)X_t$$

Now, this gives us the link between a yield of a ZCB with maturity τ at time t (so essentially the whole yield curve) and the reduced form factors X_t. How do we link yields $y_t(\tau)$ to the cascade form factors x_t that we began with?

Recall that $x_t = A(\alpha)X_t + \bar{\mu}$, which equivalently can be rewritten as $X_t = A^{-1}(\alpha)(x_t - \bar{\mu})$. We substitute this into the formula:

$$y_t(\tau) = \mu - C(\tau,\alpha,\sigma) + B(\tau,\alpha)[A^{-1}(\alpha)(x_t - \bar{\mu})]$$

$$y_t(\tau) = \mu\big(1 - \mathbf{1}'B(\tau,\alpha)A^{-1}(\alpha)\big) - C(\tau,\alpha,\sigma) + B(\tau,\alpha)A^{-1}(\alpha)x_t$$

$$y_t(\tau) = \mu\big(1 - \mathbf{1}'\gamma(\tau,\alpha)\big) - C(\tau,\alpha,\sigma) + \gamma(\tau,\alpha)x_t$$

The book calls $\gamma(\tau,\alpha) := B(\tau,\alpha)A^{-1}(\alpha)$ the vector of factor loadings. In other words, they express the partial derivative of $y_t(\tau)$ to each of the cascade form factors.

We can also obtain an expression for the forward rate at time t for a forward between times τ and $\tau + \tau'$:

$$f_t(\tau) = \mu\big(1 - \mathbf{1}'\gamma^*(\tau,\alpha,\tau')\big) + \gamma^*(\tau,\alpha,\tau')x_t - C^*(\tau,\alpha,\sigma,\tau')$$

Where we have defined factor loadings and a convexity term for forwards as:

$$\gamma^*(\tau,\alpha,\tau') = \frac{1}{\tau'}\cdot[B(\tau+\tau',\alpha) - B(\tau,\alpha)]A(\alpha)^{-1}$$

$$C^*(\tau,\alpha,\sigma,\tau') = \frac{1}{\tau'}\cdot[C(\tau+\tau',\alpha,\sigma) - C(\tau,\alpha,\sigma)]$$

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Calibration of the model

Tuckman assumes that r_t is the observed fed funds rate at t. It can be shown that the first factor loading (the loading to the short rate) depends only on α_r.

It can be shown that $A^{-1}(\alpha)$ takes the form:

$$A^{-1}(\alpha) = \begin{pmatrix} 1 & -\frac{\alpha_r}{\alpha_r - \alpha_m} & \frac{\alpha_r \alpha_m}{(\alpha_r - \alpha_m)(\alpha_r - \alpha_l)} \\ 0 & \frac{\alpha_r}{\alpha_r - \alpha_m} & \frac{\alpha_r \alpha_m}{(\alpha_r - \alpha_m)(\alpha_m - \alpha_l)} \\ 0 & 0 & \frac{\alpha_r \alpha_m}{(\alpha_r - \alpha_l)(\alpha_m - \alpha_l)} \end{pmatrix}$$

Thus the product $\gamma(\tau, \alpha) = B(\tau, \alpha)A^{-1}(\alpha)$ has first entry equal to $B(\tau, \alpha)_1 = \frac{1-e^{-\alpha_r \tau}}{\alpha_r \tau}$. The procedure in Tuckman works as follows:

1. Take a candidate value for α_r

2. Subtract $\gamma(\tau, \alpha)_1 \cdot r_t$ from both sides of the relationship between yields and factors

$$y_t(\tau) - \gamma(\tau, \alpha)_1 r_t = \mu\big(1 - \mathbf{1}'\gamma(\tau, \alpha)\big) - C(\tau, \alpha, \sigma) + \gamma(\tau, \alpha)x_t - \gamma(\tau, \alpha)_1 r_t$$

   The equation now does not involve r_t any longer on its RHS, since

$$\gamma(\tau, \alpha)x_t - \gamma(\tau, \alpha)_1 r_t = \gamma(\tau, \alpha)_2 m_t + \gamma(\tau, \alpha)_3 l_t$$

   Now define $\tilde{y}_t(\tau) = y_t(\tau) - \gamma(\tau, \alpha)_1 r_t$ and $\tilde{x}_t = (m_t, l_t)$. Also, let's write $\tilde{\gamma}(\tau, \alpha) = (\gamma(\tau, \alpha)_2, \gamma(\tau, \alpha)_3)$. We have:

$$\tilde{y}_t(\tau) = \mu\big(1 - \mathbf{1}'\gamma(\tau, \alpha)\big) - C(\tau, \alpha, \sigma) + \tilde{\gamma}(\tau, \alpha)\tilde{x}_t$$

3. We now assume that two points on the yield curve are priced exactly by the model. For example, the 2y and 10y yields of the curve. We stack them into a vector $yb_t = \big(y_t(2), y_t(10)\big)'$. With the usual transformation, we also define:

$$\widetilde{yb}_t = \begin{bmatrix} \tilde{y}(2) \\ \tilde{y}(10) \end{bmatrix} = \begin{bmatrix} y(2) - \gamma(\tau, \alpha)_1 r_t \\ y(10) - \gamma(\tau, \alpha)_1 r_t \end{bmatrix}$$

   This means we can invert the equation linking a generic yield $\tilde{y}_t(\tau)$ to $\tilde{x}_t$ to solve for $\tilde{x}_t$ based on $\widetilde{yb}_t$:

$$\begin{bmatrix} \tilde{y}_t(2) \\ \tilde{y}_t(10) \end{bmatrix} = \begin{bmatrix} \mu(1 - \mathbf{1}'\gamma(2, \alpha) \\ \mu(1 - \mathbf{1}'\gamma(10, \alpha) \end{bmatrix} - \begin{bmatrix} C(2, \alpha, \sigma) \\ C(10, \alpha, \sigma) \end{bmatrix} + \begin{pmatrix} \gamma(2, \alpha)_2 & \gamma(2, \alpha)_3 \\ \gamma(10, \alpha)_2 & \gamma(10, \alpha)_3 \end{pmatrix} \begin{bmatrix} m_t \\ l_t \end{bmatrix}$$

   If we take the first difference of both sides of the equation, we get:

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$$\begin{bmatrix} \Delta\tilde{y}_t(2) \\ \Delta\tilde{y}_t(10) \end{bmatrix} = \underbrace{\begin{pmatrix} \gamma(2,\alpha)_2 & \gamma(2,\alpha)_3 \\ \gamma(10,\alpha)_2 & \gamma(10,\alpha)_3 \end{pmatrix}}_{Y_b(\alpha)} \begin{bmatrix} \Delta m_t \\ \Delta l_t \end{bmatrix}$$

Which then gives us $\Delta\tilde{x}_t = Y_b(\alpha)^{-1}\Delta\widetilde{yb}_t$. We use this to relate the first difference of any yields in terms of the first difference of the 2y and 10y.

$$\Delta\tilde{y}_t(\tau) = \tilde{\gamma}(\tau,\alpha)\Delta\tilde{x}_t$$

$$\Delta\tilde{y}_t(\tau) = [\tilde{\gamma}(\tau,\alpha)Y_b(\alpha)^{-1}]\Delta\widetilde{yb}_t$$

In the first stage, we estimate $\beta(\tau) = \tilde{\gamma}(\tau,\alpha)Y_b(\alpha)^{-1}$ with an OLS regression for each maturity τ.

$$\hat{\beta}(\tau) = \left(\Delta\widetilde{yb}_t'\Delta\widetilde{yb}_t\right)^{-1}\Delta\widetilde{yb}_t'\Delta y$$

Then, as $\tilde{\gamma}(\tau,\alpha)Y_b(\alpha)^{-1}$ is a nonlinear function of α, we pick α solving:

$$\min_{\alpha}\sum_{\tau}\left|\left|\tilde{\gamma}(\tau,\alpha)Y_b(\alpha)^{-1} - \hat{\beta}(\tau)\right|\right|^2$$

4. Once α is estimated, we use the estimate $\hat{\alpha}$ to estimate σ minimizing the distance between model implied yield volatilities and realized volatilities of constant-maturity yields.

   According to the model, the benchmark yields $\Delta\widetilde{yb}_t$ should have varcov equal to:

$$V_t\left(\Delta\widetilde{yb}_t\right) = Y_b(\alpha)V_t(\Delta\tilde{x}_t)Y_b(\alpha)'$$

   We call $\widetilde{\Omega}(\sigma)$ the diffusion matrix of (m_t, l_t), taking the same form as $\Omega(\alpha)$ but without the zeros in the entries relating to r_t. This gives:

$$V_t(\Delta\tilde{x}_t) = \widetilde{\Omega}(\sigma)\widetilde{\Omega}(\sigma)'\Delta t = \begin{pmatrix} \sigma_m^2 & \rho\sigma_m\sigma_l \\ \rho\sigma_m\sigma_l & \sigma_l^2 \end{pmatrix}\Delta t$$

   So the model-implied variance of changes in benchmark yields shall be:

$$V_t\left(\Delta\widetilde{yb}_t\right) = \left[Y_b(\alpha)\widetilde{\Omega}(\sigma)\widetilde{\Omega}(\sigma)'Y_b(\alpha)'\right]\Delta t$$

   Using this, we can compute the model-implied variance of $\Delta\tilde{y}_t(\tau)$ for a generic τ:

$$V_t\left(\Delta\tilde{y}_t(\tau)\right) = [\tilde{\gamma}(\tau,\alpha)Y_b(\alpha)^{-1}]V_t\left(\Delta\widetilde{yb}_t\right)[\tilde{\gamma}(\tau,\alpha)Y_b(\alpha)^{-1}]'$$

$$V_t\left(\Delta\tilde{y}_t(\tau)\right) = \tilde{\gamma}(\tau,\alpha)\underbrace{Y_b(\alpha)^{-1}Y_b(\alpha)}_{I}\cdot\widetilde{\Omega}(\sigma)\widetilde{\Omega}(\sigma)'\cdot\underbrace{Y_b(\alpha)'(Y_b(\alpha)^{-1})'}_{I}\tilde{\gamma}(\tau,\alpha)'$$

$$V_t\left(\Delta\tilde{y}_t(\tau)\right) = \tilde{\gamma}(\tau,\alpha)\widetilde{\Omega}(\sigma)\widetilde{\Omega}(\sigma)'\tilde{\gamma}(\tau,\alpha)'$$

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We shall compute the difference between the varcov of all $\Delta\tilde{y}_t(\tau)$ for all τ, and the sample varcov of all yields. Note that the model implied varcov is calculated with our previous estimate of α.

$$\hat{\sigma} = \arg\min_{\sigma} \sum_{\tau} \left|\left|\tilde{\gamma}(\tau,\hat{\alpha})\tilde{\Omega}(\sigma)\tilde{\Omega}(\sigma)'\tilde{\gamma}(\tau,\hat{\alpha})' - \widehat{\mathrm{Var}}\big(\Delta y(\tau)\big)\right|\right|^2$$

5. Lastly, we pick μ to minimize squared yield errors in time across all maturities. How? Given our estimates of all other parameters, and a candidate value for μ, we can obtain the latent factors (m_t, l_t) inverting the equation that links them to the 2y and 10y rates:

$$\begin{pmatrix} \gamma(2,\alpha)_2 & \gamma(2,\alpha)_3 \\ \gamma(10,\alpha)_2 & \gamma(10,\alpha)_3 \end{pmatrix}^{-1} \left(\begin{bmatrix} \tilde{y}_t(2) \\ \tilde{y}_t(10) \end{bmatrix} - \begin{bmatrix} \mu(1-\mathbf{1}'\gamma(2,\alpha) \\ \mu(1-\mathbf{1}'\gamma(10,\alpha) \end{bmatrix} + \begin{bmatrix} C(2,\alpha,\sigma) \\ C(10,\alpha,\sigma) \end{bmatrix} \right) = \begin{bmatrix} m_t \\ l_t \end{bmatrix}$$

Or otherwise by inverting the equation linking the latent factors to two forward rates, say the 2-year and 10-year forward 1y rates. Recall that the 2-year and 10-year forward 1-year rates are linked to x_t via:

$$f_t(2) = \mu\big(1-\mathbf{1}'\gamma^*(2,\alpha,1)\big) + \gamma^*(2,\alpha,1)x_t - C^*(2,\alpha,\sigma,1)$$

$$f_t(10) = \mu\big(1-\mathbf{1}'\gamma^*(10,\alpha,1)\big) + \gamma^*(10,\alpha,1)x_t - C^*(10,\alpha,\sigma,1)$$

Which means that we can solve for x_t with:

$$\begin{bmatrix} f_t(2) \\ f_t(10) \end{bmatrix} - \begin{bmatrix} \gamma^*(2,\alpha,1)_1 r_t \\ \gamma^*(2,\alpha,1)_1 r_t \end{bmatrix} = \begin{bmatrix} \mu\big(1-\mathbf{1}'\gamma^*(2,\alpha,1)\big) \\ \mu\big(1-\mathbf{1}'\gamma^*(10,\alpha,1)\big) \end{bmatrix} + \begin{bmatrix} \gamma^*(2,\alpha,1)_2 & \gamma^*(2,\alpha,1)_3 \\ \gamma^*(10,\alpha,1)_2 & \gamma^*(10,\alpha,1)_3 \end{bmatrix} \begin{bmatrix} m_t \\ l_t \end{bmatrix} - \begin{bmatrix} C^*(2,\alpha,\sigma,1) \\ C^*(10,\alpha,\sigma,1) \end{bmatrix}$$

Adopting the same notation that we used when subtracting the contribution of the short rate from zero coupon rates, so that $\tilde{f}_t(\tau) = f_t(\tau) - \gamma^*(\tau,\alpha,\tau')_1 r_t$, we have:

$$\begin{bmatrix} \gamma^*(2,\alpha,1)_2 & \gamma^*(2,\alpha,1)_3 \\ \gamma^*(10,\alpha,1)_2 & \gamma^*(10,\alpha,1)_3 \end{bmatrix}^{-1} \left(\begin{bmatrix} \tilde{f}_t(2) \\ \tilde{f}_t(10) \end{bmatrix} - \begin{bmatrix} \mu\big(1-\mathbf{1}'\gamma^*(2,\alpha,1)\big) \\ \mu\big(1-\mathbf{1}'\gamma^*(10,\alpha,1)\big) \end{bmatrix} + \begin{bmatrix} C^*(2,\alpha,\sigma,1) \\ C^*(10,\alpha,\sigma,1) \end{bmatrix} \right) = \begin{bmatrix} m_t \\ l_t \end{bmatrix}$$

The difference between the two methods is that inverting the equation with forward rates gives a clearer meaning to m_t and l_t. They're factors tracking the risk-neutral expected path of the short rate.

Once we've done that, we use the fitted factors to price all yields in conjunction with the candidate value for μ and the other six estimated parameters. This allows us to compute a difference between model-implied and actual yields, which we use to pick the best $\hat{\mu}$:

$$\hat{\mu} = \arg\min_{\mu} \sum_{t=1}^{T} \left|\left|Y_t - y_t\right|\right|$$

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Where Y_t is the vector of model implied yields across all maturities, and y_t is the vector of actual yields across all maturities, both at a given time t.

Deriving risk premia

We assume only the long factor earns a risk premium. Define $P(t,T)$ to be the price of a bond with maturity T at time t. We're going to derive its dynamics under Q. We know that under Q, its expected return over a very small time horizon dt shall be $r_t dt$, because every discounted price is a Q-martingale. For what concerns its diffusion, we will derive it starting from the very first source of risk: the yield. Recall the equation:

$$y(t,T) = \mu\big(1 - \mathbf{1}'\gamma(T-t)\big) - C(T-t) + \gamma(T-t)x_t$$

We omit dependence on model parameters for ease of writing. Now, since $y(t,T)$ is affine in x_t:

$$dy(t,T) = [\mu_y]dt + \gamma(T-t)\Omega dW_t$$

We denote by μ_y the drift term for $y(t,T)$, which we're not interested in calculating for now. The log price is an affine function of the yield, since $p(t,T) \coloneqq \log P(t,T) = -(T-t)y(t,T)$. Hence:

$$dp(t,T) = [\mu_p]dt + [-(T-t)\gamma(T-t)\Omega]dW_t$$

Once again, we don't derive μ_p for now since we're only interested in diffusions for now. Now, the price of a bond $P(t,T)$ is an exponential function of $p(t,T)$, so:

$$dP(t,T) = \left[\frac{\partial P(t,T)}{\partial t} + \frac{\partial P(t,T)}{\partial p}\mu_p + \frac{1}{2}\frac{\partial^2 P(t,T)}{\partial p^2}(-(T-t)\gamma(T-t)\Omega)\cdot(-(T-t)\gamma(T-t)\Omega)'\right]dt + \left[\frac{\partial P(t,T)}{\partial p}(-(T-t)\gamma(T-t)\Omega)\right]dW_t$$

And since:

$$\frac{\partial P(t,T)}{\partial t} = 0 \qquad \frac{\partial P(t,T)}{\partial p} = \frac{\partial^2 P(t,T)}{\partial p^2} = P$$

$$\frac{dP(t,T)}{P} = \left[\mu_p + \frac{1}{2}(T-t)^2\gamma(T-t)\Omega\Omega'\gamma(T-t)'\right]dt + [-(T-t)\gamma(T-t)\Omega]dW_t$$

But as we mentioned in the beginning we must have $E_t\left[\frac{dP(t,T)}{P}\right] = r_t dt$. So then:

$$\frac{dP(t,T)}{P} = r_t dt - (T-t)\gamma(T-t)\Omega dW_t$$

By Girsanov, if we switch to the P measure, we would define a brownian motion W_t^P with:

$$dW_t^P = dW_t + \boldsymbol{\lambda}_t dt$$

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Where $\boldsymbol{\lambda}_t$ is a 3-item vector of prices of risk, one per factor. We assume that only the long factor earns a risk premium. So that:

$$\boldsymbol{\lambda}_t = (0 \quad 0 \quad \lambda_t)$$

This means that under the P measure, the differential of P should look like this:

$$\frac{dP(t,T)}{P(t,T)} = \left[r_t + \underbrace{\lambda_t(T-t)\gamma(T-t,\alpha)_3\sigma_l}_{\text{compensation for risk}}\right]dt + [-(T-t)\gamma(T-t,\alpha)\Omega]dW_t^P$$

The compensation for risk is the product of the bond's duration $(T-t)$, the loading to the long factor $\gamma(T-t,\alpha)_3$, the volatility of the long factor, and a price of risk λ_t. Now define:

$$\mu_P = P[r_t + \lambda_t(T-t)\gamma(T-t,\alpha)_3\sigma_l]$$

$$\sigma_P = P[-(T-t)\gamma(T-t,\alpha)\Omega]$$

Now consider the log price of a bond, $p(t,T) := \log P(t,T)$. By Ito's lemma:

$$dp(t,T) = \left[\frac{1}{P}\mu_P - \frac{1}{2}\frac{1}{P^2}\sigma_P^2\right]dt + \left[\frac{1}{P}\sigma_P\right]dW_t^P$$

$$dp(t,T) = \left[r_t + \lambda_t(T-t)\gamma(T-t,\alpha)_3\sigma_l - \frac{1}{2}(T-t)^2\gamma(T-t,\alpha)\Omega\Omega'\gamma(T-t,\alpha)'\right]dt + [-(T-t)\gamma(T-t,\alpha)\Omega]dW_t^P$$

This will help us in a second. Consider a strategy designed as follows:

1. Buy a ZCB with maturity $\tau + \Delta\tau$ at time t
1. Sell a ZCB with maturity τ at time t

The returns of the strategy should be equal to the forward rate with tenor $\Delta\tau$ locked in at time t, minus the $\Delta\tau$ maturity yield prevailing at time τ. The strategy's returns can be written as:

$$R_t^\tau = [p(\tau,\tau+\Delta\tau) - p(t,\tau+\Delta\tau)] - [p(\tau,\tau) - p(t,\tau)]$$

Recall the definition of forward rate at time t with maturity τ and tenor $\Delta\tau$:

$$f_t(\tau) = \frac{p(t,\tau) - p(t,\tau+\Delta\tau)}{\Delta\tau}$$

We rearrange the terms in R_t^τ and get:

$$R_t^\tau = [p(t,\tau) - p(t,\tau+\Delta\tau)] - [p(\tau,\tau) - p(\tau,\tau+\Delta\tau)]$$

$$R_t^\tau = [f_t(\tau) - f_\tau(\tau)]\Delta\tau$$

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What $E^P[R_t^\tau]$? Well, we have a form for dp and we can integrate over time.

$$E_t^P[R_t^\tau] = E_t\left[\int_t^\tau dp(u,\tau+\Delta\tau) - \int_t^\tau dp(u,\tau)\right]$$

$$E_t^P[R_t^\tau] = E_t\left[\int_t^\tau dp(u,\tau+\Delta\tau)\right] - E_t\left[\int_t^\tau dp(u,\tau)\right]$$

To evaluate the two expectations we only care about the drift summand of each of the two differentials, as the diffusions have expectation equal to zero.

$$\begin{aligned} E_t^P[R_t^\tau] = \int_t^\tau & r_u + \lambda_u(\tau+\Delta\tau-u)\gamma(\tau+\Delta\tau-u,\alpha)_3\sigma_l \\ & -\frac{1}{2}(\tau+\Delta\tau-u)^2\gamma(\tau+\Delta\tau-u,\alpha)\Omega\Omega'\gamma(\tau+\Delta\tau-u,\alpha)'du \\ & -\int_t^\tau r_u + \lambda_u(\tau-u)\gamma(\tau-u,\alpha)_3\sigma_l - \frac{1}{2}(\tau-u)^2\gamma(\tau-u,\alpha)\Omega\Omega'\gamma(\tau-u,\alpha)'du \end{aligned}$$

Tuckman assumes $E_t(\lambda_u) \approx \lambda_t$. So we treat it as constant. To simplify this integral, define:

$$h(z) = \lambda_t z\gamma(z,\alpha)_3\sigma_l - \frac{1}{2}z^2\gamma(z,\alpha)\Omega\Omega'\gamma(z,\alpha)'$$

And then

$$E_t^P[R_t^\tau] = \int_t^\tau r_u + h(\tau+\Delta\tau-u)du - \int_t^\tau r_u + h(\tau-u)du$$

$$E_t^P[R_t^\tau] = \int_t^\tau [h(\tau+\Delta\tau-u) - h(\tau-u)]\,du$$

Now perform a change of variable $s = \tau - u$:

$$E_t^P[R_t^\tau] = \int_0^{\tau-t} [h(s+\Delta\tau) - h(s)]\,ds$$

We split the integral again:

$$E_t^P[R_t^\tau] = \int_0^{\tau-t} h(s+\Delta\tau)\,ds - \int_0^{\tau-t} h(s)\,ds$$

$$E_t^P[R_t^\tau] = \int_{\Delta\tau}^{\tau+\Delta\tau-t} h(u)\,du - \int_0^{\tau-t} h(u)\,du$$

Which one may rewrite as:

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$$E_t^P[R_t^\tau] = \int_{\tau-t}^{\tau+\Delta\tau-t} h(u)\,du - \int_0^{\Delta\tau} h(u)\,du$$

Tuckman now assumes that the contribution of the second term is approximately zero for very short u. Hence the expected return is entirely given by:

$$E_t^P[R_t^\tau] = \int_{\tau-t}^{\tau+\Delta\tau-t} h(u)\,du$$

And with a last change of variable $u = t + \Delta\tau - t - s$ for $s \in [0, \Delta\tau]$:

$$E_t^P[R_t^\tau] = \int_0^{\Delta\tau} h(\tau + \Delta\tau - t - s)\,ds$$

$$E_t^P[R_t^\tau] = \lambda_t \overbrace{\int_0^{\Delta\tau} [(\tau + \Delta\tau - t - s)\gamma(\tau + \Delta\tau - t - s, \alpha)_3\sigma_l\,ds}^{RP_d(t,\tau,\Delta\tau)}$$
$$\underbrace{- \int_0^{\Delta\tau} \frac{1}{2}(\tau + \Delta\tau - t - s)^2\gamma(\tau + \Delta\tau - t - s, \alpha)\Omega\Omega'\gamma(\tau + \Delta\tau - t - s, \alpha)']\,ds}_{RP_c(t,\tau,\Delta\tau)}$$

Using these two amount of risk terms we get:

$$E_t^P[R_t^\tau] = \lambda_t RP_d(t, \tau, \Delta\tau) + RP_c(t, \tau, \Delta\tau)$$

So we can tie it all together with the forward rates:

$$E_t^P[R_t^\tau] = E_t^P[f_t(\tau) - f_\tau(\tau)]\Delta\tau$$

$$E_t^P[R_t^\tau] = (f_t(\tau) - E_t^P[f_\tau(\tau)])\Delta\tau$$

The forward rate at τ for a very short horizon $\Delta\tau \to 0$ can be approximated with the short rate r_τ. Why? Here is a simple argument:

$$f_t(\tau) = \frac{p(t,\tau) - p(t, \tau + \Delta\tau)}{\Delta\tau} = -\frac{p(\tau, \tau + \Delta\tau)}{\Delta\tau}$$

$$f_t(\tau) = -\frac{\log E_\tau\left[e^{-\int_\tau^{\tau+\Delta\tau} r_s ds}\right]}{\Delta\tau}$$

If we expand the integral to first order:

$$e^{-\int_\tau^{\tau+\Delta\tau} r_s ds} = 1 - r_\tau\Delta\tau + o(\Delta\tau)$$

So then

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$$f_t(\tau) = -\frac{1}{\Delta\tau}\log E_\tau[1 - r_\tau\Delta\tau + o(\Delta\tau)]$$

Since everything in the expectation is now τ-measurable:

$$f_t(\tau) = -\frac{1}{\Delta\tau}\log[1 - r_\tau\Delta\tau + o(\Delta\tau)]$$

And by a first-order expansion of $\log(1+x)$:

$$f_t(\tau) = \frac{r_\tau\Delta\tau + o(\Delta\tau)}{\Delta\tau} = r_\tau + o(1)$$

Which explains why $f_t(\tau) \to r_\tau$ as $\Delta\tau \to 0$. We now use this fact to write:

$$\lambda_t RP_d(t,\tau,\Delta\tau) + RP_c(t,\tau,\Delta\tau) = [f_t(\tau) - E_t^P[r_\tau]]\Delta\tau$$

In order to identify λ_t, we need to choose two maturities τ, τ' that fulfill this condition.

$$\frac{\lambda_t RP_d(t,\tau,\Delta\tau) + RP_c(t,\tau,\Delta\tau)}{\Delta\tau} = [f_t(\tau) - E_t^P[r_\tau]]$$

$$\frac{\lambda_t RP_d(t,\tau',\Delta\tau) + RP_c(t,\tau',\Delta\tau)}{\Delta\tau} = [f_t(\tau') - E_t^P[r_{\tau'}]]$$

$$\frac{\lambda_t[RP(t,\tau',\Delta\tau) - RP(t,\tau,\Delta\tau)] + [RP_c(t,\tau',\Delta\tau) - RP_c(t,\tau,\Delta\tau)]}{\Delta\tau} = [f_t(\tau') - f_t(\tau)] - [E_t^P(r_{\tau'}) - E_t^P(r_\tau)]$$

For very large τ, τ', Tuckman simplifies $E_t^P(r_\tau) - E_t^P(r_{\tau'})$ by the properties of mean reversion of the short rate. Hence:

$$\lambda_t = \frac{[f_t(\tau') - f_t(\tau)] + [RP_c(t,\tau',\Delta\tau) - RP_c(t,\tau,\Delta\tau)]}{\frac{1}{\Delta\tau}[RP_d(t,\tau',\Delta\tau) - RP_d(t,\tau,\Delta\tau)]}$$

And if we can make than estimate for $\hat{\lambda}_t$, then we can compute the real expectation of $E_t^P(r_\tau)$ as:

$$E_t^P(r_\tau) = f_t(\tau) - \frac{\hat{\lambda}_t RP_d(t,\tau,\Delta\tau) + RP_c(t,\tau,\Delta\tau)}{\Delta\tau}$$

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